Power BI Dashboard Documentation

Project Title: Hotels Reservations Dashboard

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Tool: Microsoft Power BI Desktop

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1. Project Overview

The Hotels Reservations Dashboard provides comprehensive insights into hotel booking patterns, customer behavior, and revenue performance for both City and Resort hotels. The dashboard helps management monitor key metrics such as Total Booking, revenue, cancellations, and customer segmentation.

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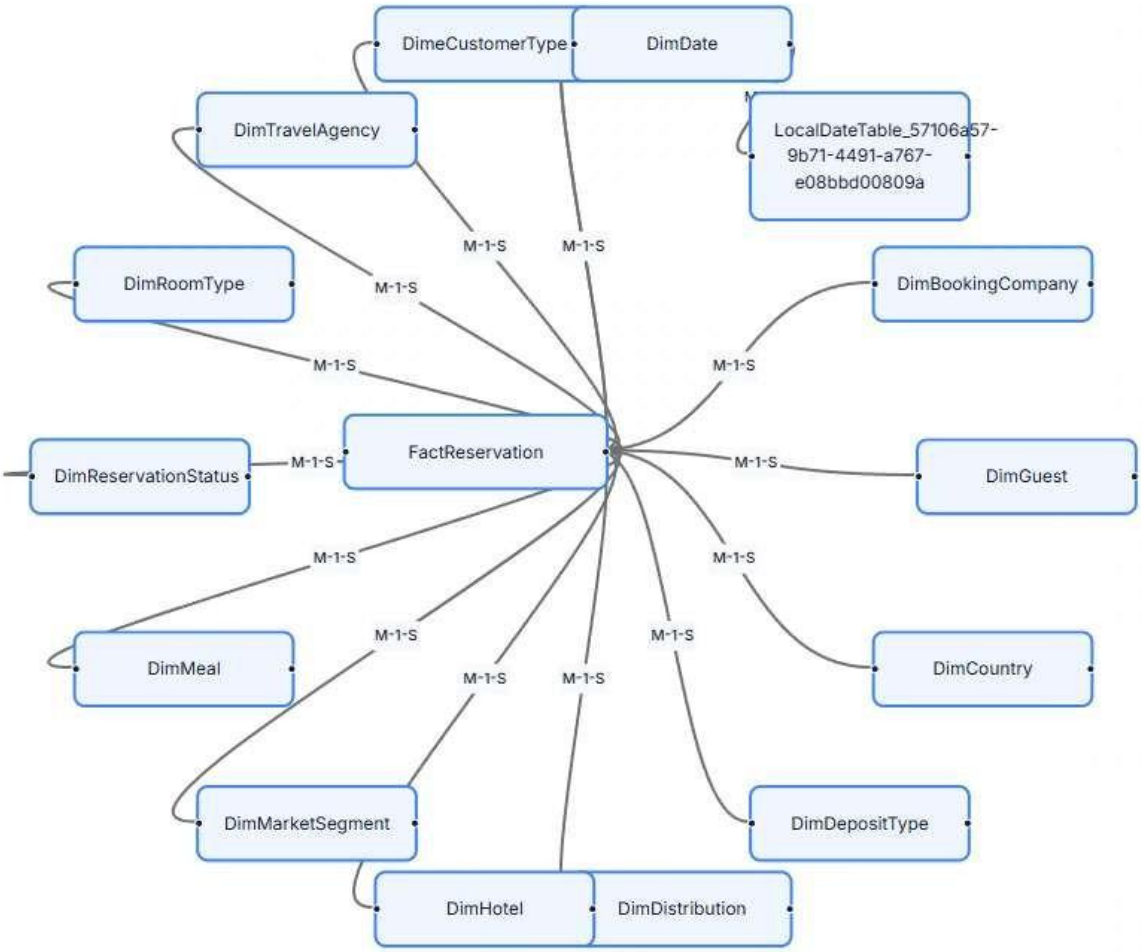
2. Data Preparation

The dataset was sourced from Kaggle (Hotel Booking Dataset). Data was loaded into Power BI and cleaned using Power Query Editor. Missing values were handled, data types corrected, and new columns were created to enhance analysis.

Step	Description
Data Import	Loaded hotels.csv, & countrycodes.csv into Power BI.
Cleaning	Removed nulls, duplicates, and adjusted date types.
Transformations	Added columns: Revenue, Length of Stay, Peak Season, Stay Type.
Measures	Created DAX measures for KPIs such as Total Bookings, Revenue, ADR, Occupancy Rate, and Cancellation Rate.

3. Data Model

A star schema was designed with one fact table (Hotels) and supporting dimension tables for Date, Market Segment, and Customer Type. Relationships were defined as one-to-many from dimension to fact tables.



4. Dashboard Design

Page 1: Executive Overview

KPIs: - Total Bookings - Total Revenue - ADR (Average Daily Rate) – Number of Nationalities - Cancellation Rate

Visuals: - Date Dimensionally Booking Trends (Line Chart)
- Total Revenue by Hotel Type (City vs Resort) (Donut Chart)
- Total Booking by Market Segment (Column Chart)
- Top 5 Countries by Revenue (Bar Chart)
- Interactive slicers: Date, Hotel Type, Market Segment

Page 2: Guest Behavior

KPIs: - Total Special Requests – Average Length of stay – Repeated Guest Rate – Average Revenue per Booking

Visuals: - Stay Type Distribution (Donut Chart)
- Total Booking by Meal Type (Bar Chart)
- Bookings Behavior by Guest (Decomposition Tree)
- Top 5 Guest with the most booking
- Interactive slicers: Date, Hotel Type, Market Segment, Room Type, Customer & Stay Type.

Page 3: Hotel Performance (City vs Resort)

KPIs: - Total Revenue – City Hotel Revenue – Resort Hotel Revenue – City Hotel Cancellation Rate - Resort Hotel Cancellation Rate.

Visuals: - ADR (Average Daily Rate) by Hotel Type (Donut Chart)
- Date Dimensionally Revenue Trends by Hotel Type (Line Chart)
- Cancellation by Hotel and Market Segment (Stacked Column Chart)
- Revenue by peak Season and Hotel Type (Clustered Column Chart)
- Question & Answer Chart
- Interactive slicers: Date, Hotel Type, Market Segment

Drillthrough page for Country Details

KPIs: - Total Bookings - Total Revenue – Repeated Guest Rate – Average Daily Rate

Visuals: - Market Segment by Month shows Revenue & Total Booking (Matrix Table)
- Total Booking by Hotel Type (Donut Chart)
- Bookings by assigned room type (Column Chart)
- Date Dimensionally Booking Trends
- Cancellation rate (Gauge Chart)

Tooltip page for Customer Type

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5. Interactivity Features

The dashboard includes multiple interactive features such as slicers for filtering, drillthrough for detailed country analysis, and tooltip pages for dynamic insight display. These enhance user exploration and decision-making efficiency.

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6. Key Insights

- City Hotels account for around 60% of total bookings.
- Revenue peaks during mid-year months, coinciding with vacation seasons and peak demand.
- Resort Hotels show higher cancellation rates (~30%) compared to City Hotels (~22%).
- The ADR remains higher for Resort Hotels, suggesting premium pricing strategies driven by leisure travel and holiday demand.
- A wide diversity of nationalities indicates a healthy mix of international guests.
- The *Online TA (Travel Agent)* segment dominates bookings, followed by *Offline TA/TO* and *Direct bookings*, suggesting that online platforms are the primary sales channel.

- The highest revenues are generated by guests in countries with strong tourism ties or business travel connections to the hotel's location.
- The average stay is short (1–3 nights), typical for city visitors.
- The repeated guest rate is modest, indicating potential to improve loyalty programs
- Bed & Breakfast" is the most common meal plan, indicating a strong preference for simple, inclusive offerings.
- The decomposition tree shows that *market segment* and *customer type* are strong determinants of booking behavior and revenue.
- A small number of guests account for a large portion of bookings — a potential target group for loyalty initiatives.
- City Hotels generate higher overall revenue due to higher booking volumes.
- Both hotel types see revenue surges during summer months, confirming the seasonality effect.
- Resort Hotels' revenue spikes more significantly in peak season, aligning with tourism cycles.